

Project IntelliScrap AI

Empowering Informal Recyclers with Multimodal Edge Intelligence

Event: Build with Gemma Hackathon — ABU Zaria

Developers: Team Nexus

Target Track: Track 1: Gemma for Local Languages & Literacy

Date: July 2026

Target Demographic: Informal Waste Pickers (*Baban Bola*) — Zaria, Northern Nigeria

1. EXECUTIVE SUMMARY

In developing urban and semi-urban hubs like Zaria, informal waste pickers (locally known as *Baban Bola*) form the structural backbone of the recycling ecosystem. However, they operate in highly hazardous conditions, regularly handling toxic electronic waste (e-waste) without protection, and are systematically underpaid due to a severe lack of direct access to scrap market pricing.

IntelliScrap AI is an offline-first, mobile-optimized Progressive Web Application (PWA) designed to transform informal waste collection into a safe, structured, and financially viable micro-enterprise. Powered by the advanced native multimodal and edge-processing capabilities of **Google Gemma 4**, the application acts as an on-the-go safety and pricing co-pilot for informal waste collectors. By executing complex intelligence locally on the device, it bridges the gap between manual field labor and structured, safe recycling channels—requiring zero active internet connection for core operations.

2. THE PROBLEM STATEMENT

- **Severe Occupational Health Risks:** Waste pickers frequently handle hazardous items such as leaking lead-acid batteries, mercury-containing fluorescent tubes, and bloated lithium-ion cells without knowing they are toxic.
- **Information Asymmetry & Financial Exploitation:** Middlemen systematically underpay informal collectors because the latter have no means of knowing the real-time, global-to-local market value of specific scrap grades (e.g., distinguishing high-value copper from brass, or HDPE from lower-grade plastics).
- **Severe Soil & Water Contamination:** Improper handling and open burning of electronic scrap releases heavy metals directly into the local agricultural soils of Zaria, directly threatening the essential Kubanni River watershed and groundwater tables.

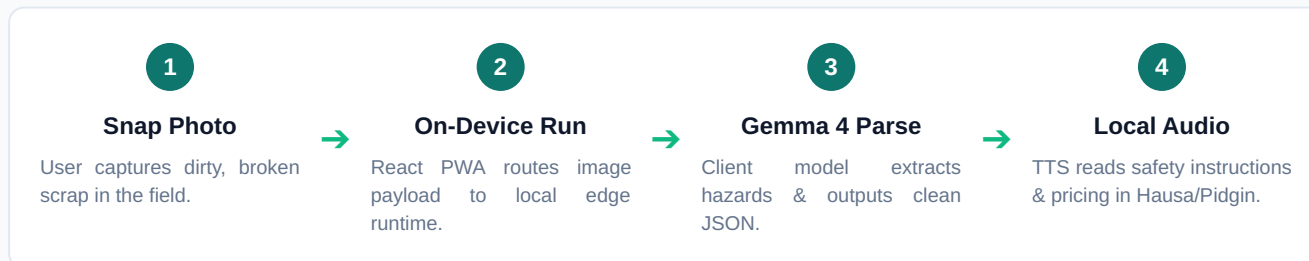
3. DESIGN THEME & HACKATHON ALIGNMENT

IntelliScrap AI directly aligns with the core directives of the **Gemma 4 Good** framework, specifically targeting:

- **Gemma for Local Languages & Literacy (Track 1):** It intentionally breaks down technical and structural barriers for an underserved, low-literacy population through native localized audio workflows in Hausa and Pidgin, combined with an intuitive, visual-first mobile interface.
- **Global Resilience & Sustainability:** It builds a practical system of tomorrow, enabling edge-based, offline environmental monitoring and toxic waste mitigation in an infrastructure-constrained environment.

4. THE SOLUTION & APP WORKFLOW

InteliScrap AI addresses these systemic challenges through an intuitive, visual-first, **offline-first workflow**. The system translates raw, complex physical objects directly into localized safety recommendations and optimal market value metrics without hitting external APIs during field operations.



Key Application Features

- **Multimodal Edge Material Classifier:** Built on Gemma 4's native multimodal capabilities, allowing the model to process dusty, dirty, or partially broken real-world materials locally and categorize them accurately (e.g., *"Urea Fertilizer bag (Polypropylene)"* or *"Transformer Winding (Copper)"*).
- **Deterministic JSON Safety Assistant:** Gemma 4 parses the visual scene to extract precise safety risk data in structured JSON format. The app instantly renders non-technical handling warnings (e.g., *"Warning: Contains Lead. Do not break open. Avoid direct skin contact."*).
- **Localized Price Indexing & Token Translation:** The app cross-references the classified material against a locally cached commodity price index table to calculate fair market payouts in Nigerian Naira (NGN) based on volume estimation.
- **Low-Literacy Multilingual Support:** Incorporates lightweight Text-to-Speech (TTS) technology to read aloud findings and safety directions in Hausa and Pidgin, ensuring the application remains fully accessible to all waste pickers regardless of reading ability.

5. TECHNICAL ARCHITECTURE & TECH STACK

LAYER	TECHNOLOGY	PURPOSE & RATIONALE
Frontend	React (Progressive Web App)	Provides a responsive mobile interface with aggressive service-worker caching, allowing the app to launch and render fully offline.
Edge AI Engine	Google Gemma 4 (Quantized E2B/ E4B)	Runs client-side in the browser via WebLLM / MediaPipe LLM Inference API. Handles local vision-to-text inference and structural reasoning without consuming cellular data.
Local Storage	IndexedDB / SQLite (WASM)	Manages local application state, stores structural historical scan logs, and holds the active localized scrap commodity pricing index matrix.
Sync Tier Backend	FastAPI	Asynchronous Python framework serving as an <i>opportunistic synchronization gateway</i> . It processes batch data syncs when users enter network coverage.
Infrastructure	Railway	Hosts the central management console, centralized PostgreSQL ledger, and serves updated regional price index sheets down to the client devices during sync windows.

6. WHY GEMMA 4? (THE COMPETITIVE ADVANTAGE)

Rather than treating AI as a mere cloud-hosted chatbot, IntelliScrap AI leverages the highly distinct structural strengths of the Gemma 4 architecture:

- Native Frontier Multimodal Understanding:** Gemma 4 excels at processing visual inputs in complex, low-lighting, and real-world noisy environments. This enables accurate identification of severely degraded or dirty scrap pieces that traditional, shallow vision models misclassify.
- Highly Efficient Edge Footprint:** The optimized Gemma 4 E2B and E4B model variants are purposefully built to enable high-performance local execution on resource-constrained host devices, such as commercial smartphones or low-cost single-board edge nodes, without relying on persistent server infrastructure.
- Structured Contextual Grounding & Tool Formatting:** Gemma 4's strong native function calling and structural JSON capabilities ensure that the vision parser cleanly outputs predictable schemas (`material_class`, `hazard_level`). This prevents model hallucination and allows instant on-device validation against local storage database tables.

7. SOCIAL & ENVIRONMENTAL IMPACT

- **Direct Economic Uplift:** Puts transparent, unmanipulated pricing metrics directly into the hands of marginalized *Baban Bola*, drastically increasing their bargaining power, eliminating middleman exploitation, and growing household income.
- **Public Health Intervention:** Directly reduces rates of respiratory illness, chemical burns, and heavy metal poisoning among urban scavengers and their surrounding communities by providing actionable, native-tongue protective guidance.
- **Water & Eco-System Conservation:** Halts improper field disposal and the dangerous open-air burning of electronic scrap, mitigating hazardous chemical runoff into local water bodies, safeguarding the critical Kubanni River watershed, and preserving surrounding soils for Zaria's agricultural sustainability.

"InteliScrap AI reimagines how advanced LLMs can serve the most vulnerable and critical actors in our physical environment, proving that edge-based, multilingual AI can solve structural problems in our immediate community."